

A 64×64 Image Energy Harvesting Configurable Image Sensor

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Abstract—This paper presents a configurable image sensor with energy harvesting capabilities. The image sensor's pixels can be configured as photo-sensors for image acquisition or as tiny solar cells for energy harvesting. The proposed configurable pixel design requires four pixels and uses a N-well/P-sub photodiode which is typically regarded as a parasitic diode for energy harvesting. A chip containing a 64×64 array of configurable pixels, read-out circuitry and biasing was designed and fabricated on a standard $0.5 \mu\text{m}$ CMOS process. The pixel size is $36 \mu\text{m} \times 40.8 \mu\text{m}$ with a fill factor of 42%. Test results are presented demonstrating the dual functionality of the image sensor. Measurements show that 55 nW can be harvested for an illuminance of 500 lux at the lens plane.

I. INTRODUCTION

Many sensing applications require sensor nodes to operate in remote and isolated areas for long periods of time. In these scenarios, the sensor nodes might not have access to the electric grid to recharge their batteries. Moreover, due to their remote location, replacing the batteries of these sensors will be very costly and in some instances even prohibitive. In these cases harvesting energy from the sensor's environment is an appealing solution which could eliminate battery replacement and enable long term operation of the sensor.

Sources of ambient energy can be as diverse as solar, thermal, wind, vibrations and electro-magnetic fields. Solar is one of the most abundant sources of energy in the planet and is specially suitable in outdoor monitoring scenarios where power densities can reach up to 100 mW/cm^2 . In this work we consider harvesting energy to power an image sensor or to recharge its battery. Given that image sensors are exposed to light in order to capture images, it makes sense to harvest energy from the light that reaches the sensor. The photodiodes inside the pixels of an image sensor could be used not only to sense light but also to harvest photonic energy.

The use of photodiodes as energy harvesters in CMOS image sensors has been explored in [1]–[4,6]. In [1] an additional photodiode called the power generation photodiode (PGPd) was placed in each pixel to harvest energy. A drawback of this approach is that the PGPd occupies significant pixel area (around 30 %) while the photodiode employed for image acquisition occupies a small percentage (less than 10 %). In [2] the PGPd concept was applied to a time-to-first spike (TFS) image sensor architecture. In the TFS image sensor, each pixel has two photodiodes: the PGPd for energy harvesting and a photodiode for image capture (Pd). When the TFS pixels fire, they are re-configured so that the Pd is connected in

parallel with the PGPd to increase energy collection. The PGPd photodiode in TFS image sensor still employs a fairly large percentage (about 40 %) of the pixel's area. Moreover, additional circuitry is needed inside the pixel to implement the handshaking protocol needed in a TFS sensor architecture.

The work in [3] presents a configurable energy harvesting pixel design that uses only one photodiode and seven transistors resulting in larger fill factors. In this approach, a P-diff/N-well photodiode is used to either harvest energy or to sense light. A similar pixel concept is presented in [4]. The design presented in [4], however, employs only four transistors. In both cases, a P-diff/N-well photodiode is configured to work either as an energy harvester or as a photo-sensor to acquire images. To configure the P-diff/N-well photodiode as an energy harvester, its cathode is connected to ground via an in-pixel transistor. This connection, however, activates the parasitic N-well/P-sub photodiode which diverts photo-generated charges away from the P-diff/N-well photodiode lowering the energy conversion efficiency by almost 50% [5].

The work reported in [6] addressed the problem of the parasitic N-well/P-sub photodiode by employing a DC-DC converter with a flying inductor to transfer charges from the P-diff/N-well photodiode to the load. The usage of a flying inductor allows to keep the N-well at a floating potential which turns off the parasitic N-well/P-diff photodiode. Hence, it improves the energy conversion efficiency. The pixel design in [6] includes an in-pixel one-bit memory which allows pixels to be individually configured as energy harvesters or as photo-sensors. The drawback of this pixel design is that it uses 12 transistors per pixel. Another advantage of using the flying-inductor DC-DC converter is that the inherently low photodiode voltage (0.1 to 0.5 V depending on illumination conditions) is boosted to levels that can be used to power electronic circuits and to charge a battery cell.

In this work an energy-harvesting image sensor whose pixels can be configured to work as photo-sensors to capture images or as tiny solar cells to harvest energy is presented. The proposed configurable pixel consists of four transistors and a N-well/P-sub photodiode. In previous works the N-well/P-sub photodiode was considered a parasitic diode. In this work, however, the parasitic N-well/P-sub photodiode is employed to harvest energy and to sense light. Moreover, the percentage of pixels configured to either mode (energy harvesting or photo-sensing) can be varied in discrete steps (0%, 25%, 50%, 75% and 100%) enabling the sensor to adapt to changing internal or environmental conditions. A 64×64 array of configurable

pixels along with read-out circuitry, analog-to-digital converter and bias voltage generation circuits were integrated in a single chip and fabricated on a standard $0.5\mu\text{m}$ CMOS fabrication technology. Initial tests of the image sensor chip demonstrating its sensing and energy harvesting capabilities are presented.

II. PRELIMINARIES

Fig. 1 shows the current-vs-voltage (I-V) curve of a typical photodiode under illumination and no-illumination conditions. Under illumination the I-V curve moves down by an amount equal to the photo-generated current or I_{ph} .

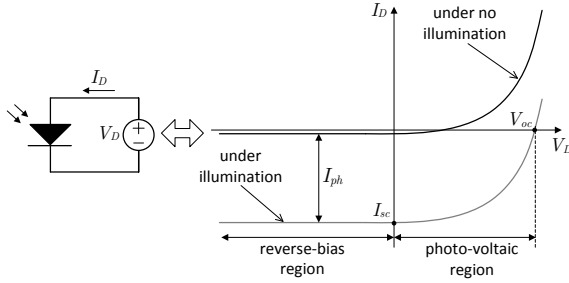


Fig. 1. Regions of operation of a photodiode under illumination and non-illumination conditions.

As it can be appreciated from the figure, two very distinctive operating regions appear when the photodiode is illuminated: the reverse-bias regions and the photo-voltaic region. In the reverse-bias region, the potential across the photodiode, V_D , is negative and the photodiode current, I_D , flows out of the diode's anode. Hence, the photodiode dissipates energy. In the photo-voltaic region, on the other hand, V_D is positive and I_D flows out of the diode's anode. Hence, the photodiode generates energy. The configurable pixel architecture exploits these two regions: it operates the photodiode in the reverse-bias region when the pixel is configured as a photo-sensor (this is the standard operating mode in CMOS image sensors). When the pixel is configured as an energy harvester, it operates the photodiode in the photo-voltaic region.

In the traditional active pixel sensor (APS) image sensor architecture, the in-pixel photodiode is reversed biased by connecting its anode to ground and its cathode to a positive reset voltage (V_{rst}) via a reset switch. Closing the reset switch charges the photodiode's depletion capacitance (C_D) to V_{rst} . When the reset switch opens, the photo-generated current (I_{ph}) discharges C_D for a fixed amount of time known as the integration time (T_{int}).

To work in the photo-voltaic region, the photodiode's cathode is connected to ground while the photodiode's anode is connected to a DC-DC converter. In [6] a H-bridge was employed to switch the photodiode's terminal according to the mode of operation. In [3] two in-pixel transistors are employed to ground the photodiode's cathode and to connect the photodiode's anode to an energy harvesting bus. In [4] an in-pixel transistor is employed to ground the cathode of the photodiode (when the pixel is in energy harvesting mode). However, additional switches outside the pixel are needed to disconnect the anode from ground and to a load or DC-DC converter.

Given that in energy harvesting mode the photodiode's cathode is connected to ground, previous approaches have used

a P-diff/N-well junction to implement the in-pixel photodiode. This photodiode choice, however, gives rise to the N-well/P-sub parasitic photodiode which lowers the energy conversion efficiency. In this work, the problem of the parasitic photodiode is solved by using the parasitic photodiode itself as a light sensor and as an energy harvester. To be able to harvest energy from a parasitic photodiode, the DC-DC converter reported in [7] is used. The proposed pixel design is described in the next section.

III. HARDWARE DESIGN

Fig. 2 shows the schematic diagram of the proposed configurable energy harvesting pixel. It consists of three NMOS transistors (M_1 , M_2 and M_H), a PMOS transistor (M_R) and a N-well/P-sub photodiode (D_1). Transistor M_R implements the reset switch.

In the sensing mode, transistor M_H is off and the pixel works as a standard APS pixel. In the APS architecture, transistor M_1 and the off-pixel current sink form a source follower amplifier. Transistor M_2 works as a switch. When M_2 is on, the source follower drives the sensing bus with a voltage proportional to V_j . The sensing bus is shared with other pixels on the same column. Only one pixel, however, drives the sensing bus at any time. To configure a pixel in the sensing mode the signal **EH enable** is set a logic low level.

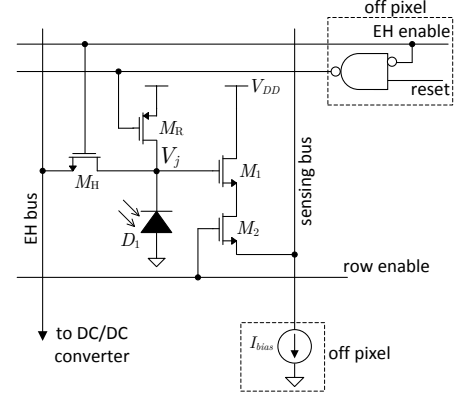


Fig. 2. Schematic diagram of the proposed configurable energy harvesting pixel.

To configure the pixel in the energy harvesting mode, the signal **EH enable** is set to a logic high level. The off-pixel NAND gate is employed to avoid turning on transistor M_R , and thus resetting the photodiode, when the pixel is working in the energy harvesting mode. When **EH enable** is logic high, transistor M_H turns on and connects the cathode terminal of the photodiode to the energy harvesting bus (EH bus). The EH bus is a global bus that connects the photodiodes of the pixels configured in energy harvesting mode to a switched-inductor DC-DC converter.

Fig. 3 shows the connection between the pixels configured in energy harvesting mode and the DC-DC converter. The DC-DC converter consists of a NMOS transistor M_S , an inductor L , a capacitor C_P and the Schottky diode D_S [7]. When the clock signal ϕ is logic high, the inductor gets charged by the photo-generated currents from the in-pixel photodiodes. When ϕ is logic low the inductor discharges into the load.

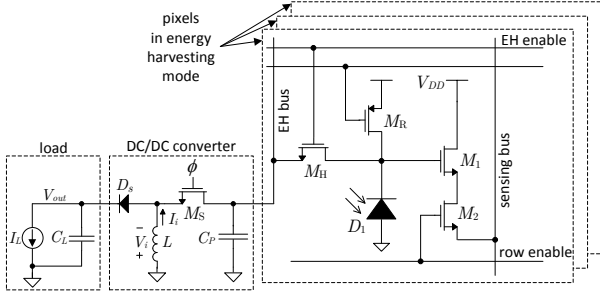


Fig. 3. Connection between the energy-harvesting pixels and the DC-DC converter.

The architecture of the configurable energy harvesting image sensor is shown in Fig. 4. The image sensor consists of an array of 64×64 configurable energy harvesting pixels, a bank of 64 correlated double sampling (CDS) units, a 64-to-1 analog multiplexer, a cyclic analog-to-digital converter (ADC), a bias-generator circuit, two vertical shift registers (to generate the reset and row enable signals) and one horizontal shift register (to generate the column enable signals).

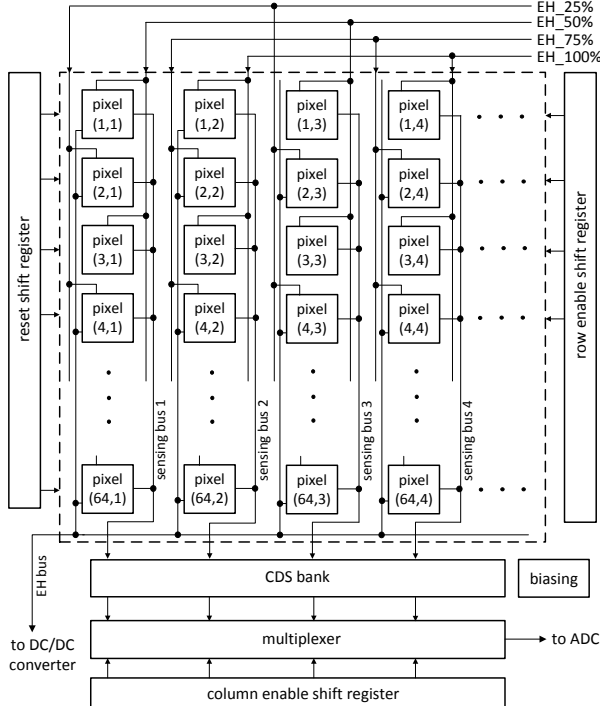


Fig. 4. Architecture of the energy harvesting image sensor.

The image sensor is read out in a raster scan fashion. The reset shift register is used to sequentially reset each row of the array. The row enable shift register enables the pixels in a row T_{int} seconds after the pixels were reset. When a pixel is enabled, it outputs (on the sensing bus) a voltage, V_{ph} , that is a function of the photo-generated current I_{ph} and T_{int} .

The image sensor has been designed so that the percentage of pixels configured in energy harvesting mode can be varied in order to respond to environmental conditions, image resolution requirement and bandwidth constraints. For instance, if the sensor's battery charge is low, more pixels can be configured to harvest energy. As more pixels are configured as energy har-

vesters, less pixels will be available for image acquisition and the image resolution will decrease. However, the missing pixel values can be estimated from the available pixels. Moreover, since fewer pixel values need to be read from the sensor, the read-out bandwidth decreases.

The percentage of pixels configured in energy harvesting mode can be varied in discrete steps (0%, 25%, 50%, 75% and 100%) using the signals **EH_25%**, **EH_50%**, **EH_75%**, and **EH_100%**. This process can be explained with the 16-pixel block spanning from pixel(1,1) to pixel(4,4) as shown in Fig. 4. The connectivity pattern in this 16-pixel block is repeated throughout the array. When **EH_25%** is logic high while **EH_50%**, **EH_75%** and **EH_100%** are logic low, 25% of pixels, that is pixel(2,1), pixel(4,1), pixel(2,3) and pixel(4,3), are configured in energy harvesting mode. To configure 50% of the pixels in energy harvesting mode, both **EH_25%** and **EH_50%** are set to logic high, while **EH_75%** and **EH_100%** are set to logic low. Similarly, to configure 75% of the pixels in energy harvesting mode, **EH_25%**, **EH_50%** and **EH_75%** are set to logic high and **EH_100%** is set to logic low. When **EH_25%**, **EH_50%**, **EH_75%** and **EH_100%** are set to logic high, 100% of the pixels are configured in energy harvesting mode. Each CDS unit in the CDS bank is a switched-capacitor circuit that outputs a voltage proportional to the difference $V_{rst} - V_{ph}$, where V_{rst} is the output of the pixel during reset. By subtracting V_{ph} from V_{rst} , the offset component of the fixed pattern noise is canceled out.

IV. RESULTS

The configurable energy harvesting image sensor was fabricated on $0.5 \mu\text{m}$ standard CMOS fabrication process. The size of the fabricated chip is $3.69 \text{ mm} \times 4.1 \text{ mm}$. The pixel area is $36 \mu\text{m} \times 40.8 \mu\text{m}$ with a fill factor of 42 %. The chip was packaged in a ceramic PGA package and mounted on a custom printed circuit board (PCB). The package chip was outfitted with a M12 lens holder and a varifocal lens. The custom PCB includes a field programmable logic array (FPGA) to generate clock and the control signals needed to acquire images and to handle serial communications with a host computer.

The test setup shown in Fig. 5 was built to characterize the image sensor. Test images were placed 70 cm away from the lens and illuminated with a 30 W white LED light source. The illuminance at the image plane was measured to be 3500 lux at the light source and 500 lux at the lens plane.

Fig. 6 shows the image *Eye* acquired under the conditions described above when 0% were configured in the energy harvesting mode, or equivalently, 100% of the pixels configured in the sensing mode. To test the energy harvesting capability of the image sensor, the percentage of pixels in energy harvesting mode was varied from 0% to 100%. Figs. 8(a), 9(a) and 10(a) show the image *Eye* acquired when 25%, 50% and 75% of the pixels are configured as energy harvesters, respectively. The energy harvesting pixels are shown as black squares and do not provide image information.

Fig. 7 shows the measured (I-V and power-vs-voltage (P-V)) from the image sensor for the image *Eye* and when 100% of the pixels are in energy harvesting mode. The figure shows that 55 nW can be harvested from the sensor for the test conditions

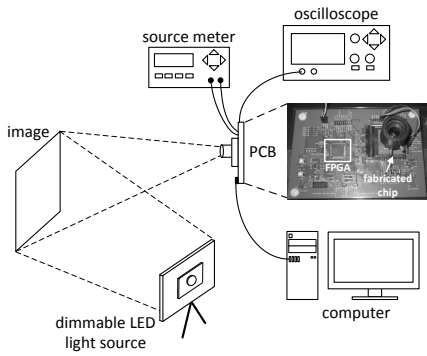


Fig. 5. Diagram of the test setup built to characterize the fabricated energy harvesting image sensor.



Fig. 6. Image Eye acquired using the fabricated configurable energy harvesting image sensor.

stated above. The I-V curve was measured with the Keithley 2401 source meter. Higher power levels can be generated by the sensor in outdoor conditions where illuminance levels can reach 100 klux. The amount of harvested energy also depends on the image content. Future tests will include characterization with different images and illumination levels.

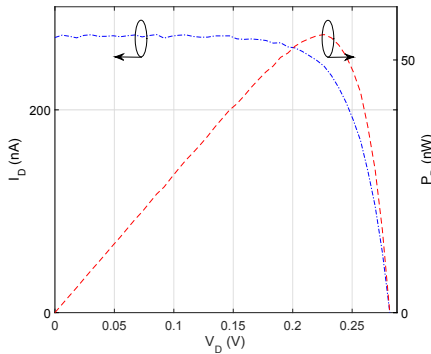


Fig. 7. I-V and P-V curves for the image Eye when 100% of pixels are in energy harvesting mode.

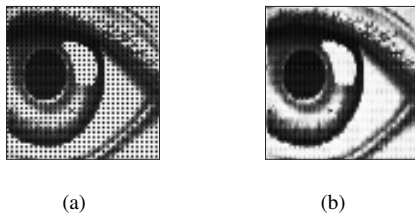


Fig. 8. Image Eye: (a) acquired when 25% of pixels are configured as energy harvesters; (b) reconstructed from available pixels.

Although the energy harvesting pixels do not provide image information, their values can be estimated from their available neighbors using image interpolation techniques. Figs. 8(b), 9(b) and 10(b) show reconstructed images when cubic interpolation was used when 25%, 50% and 75% of the pixels are configured as energy harvesters, respectively. Future work

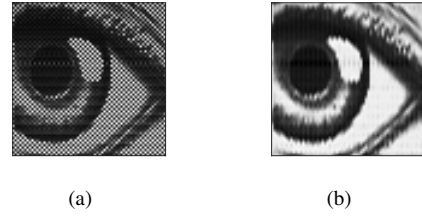


Fig. 9. Image Eye: (a) acquired when 50% of pixels are configured as energy harvesters; (b) reconstructed from available pixels.

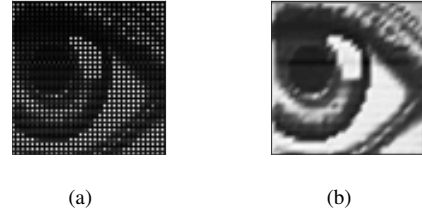


Fig. 10. Image Eye: (a) acquired when 75% of pixels are configured as energy harvesters; (b) reconstructed from available pixels.

includes adding a charger after the DC-DC converter to charge a battery or a super-capacitor with the harvested energy.

V. CONCLUSION

A configurable image sensor with energy harvesting capabilities has been presented. The image sensor's pixels can be configured as photo-sensors for image acquisition or as tiny solar cells for energy harvesting. The proposed configurable pixel design requires four pixels and, unlike previous designs, it uses a N-well/P-sub photodiode. A chip containing a 64×64 array of configurable pixels, read-out circuitry, biasing and analog-to-digital conversion was designed and fabricated on a standard $0.5 \mu\text{m}$ CMOS process. Test results are presented demonstrating the dual functionality of the sensor.

ACKNOWLEDGMENT

The authors would like to thank the National Science Foundation for its support through grant ECCS-1055169.

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